

D500

Data Visualization

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word _____	The one fact that decides _____	Answer _____	[] sure [] unsure [] guess
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Items

1.

A dashboard designer at a shipping firm with 3,000 employees must let executives compare exact quarterly revenue across eight regions as accurately as possible. The executives prefer bright colors. Which encoding supports the most accurate comparison?

- A. Position along a common aligned scale, such as bars or dots on one axis.
- B. Bubble area.
- C. Bright color intensity for each region.
- D. Slice angle in a pie chart.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

Which encoding is least accurate for judging precise differences between values?

- A. Color saturation.
- B. Position on a common scale.
- C. Length.
- D. Angle.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

An analyst at a regional airline with 40 routes has 25 charts on flight delays. She has 10 minutes with the COO, who must decide whether to add ground crew at one hub. The analysis took six weeks. What approach is best?

- A. Build up through the charts and save the conclusion for the end.
- B. Lead with the recommendation and the one or two charts that support it, highlighting the key comparison.
- C. Hand over a raw data table instead of charts.
- D. Show all 25 charts to demonstrate the six weeks of work.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

A public-health map shows each county's change in vaccination rate from -15% to $+15\%$, with zero as the meaningful midpoint. The map will be printed in a 60-page annual report. Which palette fits?

- A. A qualitative palette.
- B. A sequential palette.
- C. A diverging palette centered on zero.
- D. A rainbow palette, which is always clearest.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5. [TRANSFER — reversed stem / different variable]

A map shows population density from low to high, with no meaningful midpoint. Which palette fits?

- A. Any palette works as long as there is a legend.
- B. A qualitative palette.
- C. A diverging palette.
- D. A sequential palette.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

A KPI dashboard for a bank with 5 million customers marks on-target metrics green and off-target metrics red, using color alone. Some viewers have red-green color vision deficiency. The dashboard was built in a popular BI tool. What is the best fix?

- A. Never use color on dashboards.
- B. Switch to a different BI tool.
- C. Add a second cue, such as icons or text labels, and use a colorblind-safe palette like blue and orange.
- D. Make the red and green brighter.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7.

A slide for a software firm's quarterly review shows bars for customer satisfaction scores of 82 and 86, with the y-axis starting at 80. The second bar looks three times as tall. The CEO will present it at a conference in Chicago. What is the problem?

- A. No problem, because the axis values are labeled.
- B. The bars should be 3-D for more depth.
- C. It is only a problem when shown outside the company.
- D. The truncated axis exaggerates a 4-point (about 5%) difference; bar charts should start at zero.

Target word _____	The one fact that decides _____	Answer ____	[] sure [] unsure [] guess
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8. [TRANSFER — reversed stem / different variable]

For which chart type is a y-axis that does not start at zero usually acceptable?

- A. A pie chart.
- B. A line chart showing change over time.
- C. A bar chart.
- D. A stacked bar chart.

Target word _____	The one fact that decides _____	Answer ____	[] sure [] unsure [] guess
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9.

A marketing chart for a hotel group with 22 properties uses 3-D bars, a photo background, heavy gridlines, and drop shadows to show occupancy by month. The designer won an award last year. Which change best improves it?

- A. Add more 3-D depth and texture.
- B. Remove the non-data ink and use plain 2-D bars or a line.
- C. Switch to a pie chart.
- D. Keep it because the designer is award-winning.

Target word _____	The one fact that decides _____	Answer ____	[] sure [] unsure [] guess
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10.

A nonprofit with 18 volunteer chapters wants to show which chapters raised the most money this year, so donors can compare them. The chapters meet monthly. Which chart fits best?

- A. A pie chart with 18 slices.
- B. A line chart of the 18 chapters.
- C. A sorted horizontal bar chart.
- D. A monthly line chart of meeting counts.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

An infographic for a city transit agency with 900 buses shows daily ridership rising from 40,000 to 50,000. The bar for 50,000 is drawn twice as tall as the bar for 40,000. The graphic was made for a press release. What is the lie factor?

- A. 0.25
- B. 1.25
- C. 2
- D. 4

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

Data values rise 50%. For an accurate graphic (lie factor of 1), how tall should the new bar be compared with the old one?

- A. 1.5 times as tall
- B. 0.5 times as tall
- C. The same height
- D. 2 times as tall

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer A Concept 1

A dashboard designer at a shipping firm ~~with 3,000 employees~~ must let executives compare exact quarterly revenue across eight regions as accurately as possible. ~~The executives prefer bright colors.~~ Which encoding supports the most accurate comparison?

Target word: most accurate encoding **Decides:** Precise comparison of values.

A	Correct	Position on a common scale is judged most accurately.
B	Adjacent category	People misjudge area differences.
C	Camouflage	Answers the preference; saturation is the least accurate encoding.
D	Confusion set	Angles are harder to compare than positions.

2. Answer A Concept 1 • transfer

Which encoding is least accurate for judging precise differences between values?

Target word: least accurate — reverse **Decides:** Bottom of the encoding ranking.

A	Correct	Shading is the least precise channel.
B	Inverted operation	The most accurate channel.
C	Confusion set	Length is judged well, just below position.
D	Adjacent category	Weaker than position or length, but better than saturation.

3. Answer B Concept 2

An analyst at a regional airline ~~with 40 routes~~ has 25 charts on flight delays. She has 10 minutes with the COO, who must decide whether to add ground crew at one hub. ~~The analysis took six weeks.~~ What approach is best?

Target word: presentation approach **Decides:** One decision, 10 minutes.

A	Inverted operation	Busy decision makers need the takeaway first.
B	Correct	Shapes the story around the decision.
C	Adjacent category	Forces the COO to do the analysis.
D	Camouflage	Effort does not help the COO decide.

4. Answer C Concept 3

A public-health map shows each county's change in vaccination rate from -15% to $+15\%$, with zero as the meaningful midpoint. ~~The map will be printed in a 60-page annual report.~~ Which palette fits?

Target word: palette type **Decides:** Values on both sides of a meaningful zero.

A	Confusion set	For unordered categories.
B	Adjacent category	Hides the meaningful midpoint.
C	Correct	Two hues show direction away from the midpoint.
D	Absolute language	Rainbows distort order and print poorly.

5. Answer D Concept 3 • transfer

A map shows population density from low to high, with no meaningful midpoint. Which palette fits?

Target word: palette type — reverse case **Decides:** Ordered, one direction.

A	Absolute language	A legend does not fix a misleading palette.
B	Adjacent category	Loses the order.
C	Confusion set	Implies a midpoint that does not exist.
D	Correct	Light-to-dark shows low-to-high.

6. Answer C Concept 4

A KPI dashboard for a bank ~~with 5 million customers~~ marks on-target metrics green and off-target metrics red, using color alone. Some viewers have red-green color vision deficiency. ~~The dashboard was built in a popular BI tool.~~ What is the best fix?

Target word: accessibility fix **Decides:** Meaning carried by red vs. green alone.

A	Absolute language	Color is fine as a redundant cue.
B	Camouflage	The tool is not the cause.
C	Correct	Meaning no longer depends on color alone.
D	Dropped element	Drops the need for a non-color cue; brightness does not help.

7. Answer D Concept 5

A slide for a software firm's quarterly review shows bars for customer satisfaction scores of 82 and 86, with the y-axis starting at 80. The second bar looks three times as tall. ~~The CEO will present it at a conference in Chicago.~~ What is the problem?

Target word: design flaw **Decides:** A truncated bar axis.

A	Dropped element	Drops the zero-baseline rule; viewers read length first.
B	Adjacent category	Adds distortion instead of fixing it.
C	Camouflage	A misleading chart misleads any audience.
D	Correct	Bar length must be proportional to value.

8. Answer B Concept 5 • transfer

For which chart type is a y-axis that does not start at zero usually acceptable?

Target word: where truncation is acceptable — reverse **Decides:** Position, not length, encodes value.

A	Adjacent category	Pies have no y-axis.
B	Correct	Lines encode position, so zooming shows change.
C	Inverted operation	Bars need zero.
D	Confusion set	Still length-based; needs zero.

9. Answer B Concept 6

A marketing chart for a hotel group ~~with 22 properties~~ uses 3-D bars, a photo background, heavy gridlines, and drop shadows to show occupancy by month. ~~The designer won an award last year.~~ Which change best improves it?

Target word: improvement **Decides:** Decoration that carries no data.

A	Inverted operation	Adds more non-data ink.
B	Correct	Raises the data-ink ratio and clarity.
C	Adjacent category	A pie cannot show twelve months well.
D	Camouflage	Credentials do not fix the chart.

10. Answer C Concept 7

A nonprofit ~~with 18 volunteer chapters~~ wants to show which chapters raised the most money this year, so donors can compare them. ~~The chapters meet monthly.~~ Which chart fits best?

Target word: chart for comparison **Decides:** Comparing 18 categories.

A	Confusion set	Too many slices; angles are hard to compare.
B	Adjacent category	Lines imply order over time.
C	Correct	Bars on a common baseline, sorted, make comparison easy.
D	Camouflage	Answers the meeting detail, not fundraising.

11. Answer D Concept 8

An infographic for a city transit agency ~~with 900 buses~~ shows daily ridership rising from 40,000 to 50,000. The bar for 50,000 is drawn twice as tall as the bar for 40,000. ~~The graphic was made for a press release.~~ What is the lie factor?

Target word: lie factor **Decides:** Data change 25%; graphic change 100%.

A	Inverted operation	Divides data effect by graphic effect.
B	Dropped element	The data ratio alone.
C	Wrong stem number	The drawn ratio, not the lie factor.
D	Correct	$100\% / 25\% = 4$.

12. Answer A Concept 8 · transfer

Data values rise 50%. For an accurate graphic (lie factor of 1), how tall should the new bar be compared with the old one?

Target word: solve for the drawn size **Decides:** Graphic change must equal data change.

A	Correct	A 50% rise drawn as a 50% taller bar.
B	Inverted operation	Shrinks the bar for an increase.
C	Dropped element	Shows no change at all.
D	Confusion set	Draws a 100% rise, a lie factor of 2.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	4	
Wrong stem number	A figure copied from the prompt that is not the answer.	1	
Adjacent category	The right idea in the wrong bucket.	9	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	6	
Absolute language	Always, never, guarantee — where judgment applies.	3	
Confusion set	Every option is real; the test is exact role-matching.	7	
Camouflage	Answers a stem fact that was never the question.	6	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Ranking of visual encodings	People judge position on a common scale most accurately, then length, then angle and area, with color saturation least accurate.	Using bubble size, angle, or color for precise comparisons.	“Compare exact values as accurately as possible.”
2	Audience and message first	Start with the decision the audience must make and show only the evidence that supports it, leading with the takeaway.	Showing every chart to prove the work, or saving the conclusion for last.	Limited time with a decision maker.
3	Color palette types	Sequential for ordered low-to-high values; diverging for values around a meaningful midpoint; qualitative for unordered categories.	Using a sequential or rainbow palette for data centered on zero.	Negative and positive values, or a stated midpoint.
4	Accessible color	Never encode meaning in color alone; add labels, icons, or position and use palettes that work for color-vision deficiency.	Relying on red vs. green.	Status shown only by red and green.
5	Zero baselines for bars	Bar length encodes value, so bar charts need a zero baseline; line charts may zoom to show change.	Truncating a bar chart's axis to dramatize a small difference.	A bar axis starting well above zero.
6	Data-ink ratio	Remove ink that does not carry data (3-D effects, heavy gridlines, background images, shadows).	Adding decoration to make a chart “engaging.”	A chart described with several decorative effects.
7	Chart choice by task	Line for trends over time, bar for comparing categories, scatter for relationships, and pie only for a few parts of one whole.	Choosing a pie or a table for a comparison task.	The task verb: trend, compare, relate, part of whole.
8	Lie factor	Lie factor = size of the effect shown in the graphic ÷ size of the effect in the data; 1 is accurate.	Reading the drawn ratio itself as the lie factor.	A percentage change in data and a drawn size change.

Course scope (public description): Data Visualization covers the application of design principles, human perception, color theory, and effective storytelling in the context of data visualization. It addresses presenting data to others and advancing technology with visualization tools enabling data scientists to share their findings and support organizational decision-making processes. Additionally, this course focuses on how to visually encode and present data to an audience.